

Package ‘loco’

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Type Package

Title Functions to Calculate and Plot Local Correlations Between
Features in High-Dimensional Point-Cloud Data

Version 0.1.0

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Description Quickly detect local correlation patterns between features in point cloud data (like single-cell data in biological applications). Instead of clustering the data points and looking for correlations within every cluster

LoCo creates neighbourhoods of points, detects correlations within those neighbourhoods and filters correlations that change smoothly through the point cloud.

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LinkingTo Rcpp

Imports Rcpp,ggplot2,uwot,stats

SystemRequirements C++17

RoxygenNote 7.1.2

NeedsCompilation yes

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<code>add_umap_coords</code>	<i>Creates UMAP space for raw data and adds the UMAP-coordinates UMAP1/UMAP2 to RawData, Neighbourhoods and Correlations. run: <code>umapAddedLocoResult <- add_umap_coords(locoResult)</code> to add the UMAP coords to the loco result. <code>umapAddedLocoResult</code> can be used to plot the correlations in neighbourhoods on a UMAP. Details: To plot local correlations across neighbourhoods LoCo needs a space to plot those correlations in. One possibility is to create the UMAP space of the raw data. Within this UMAP space you can inspect original raw features and additionally assign UMAP coordinates to neighbourhoods by taking the central cell of all neighbourhoods (the anchor cell) and assigning the UMAP coordinates of this cell to their neighbourhood. After running <code>add_umap_coords()</code> you can run <code>plot_local_correlation_map()</code> to plot local correlations within this UMAP space. You do not have to use these umap coordinates but this function makes it easy to create the UMAP space for the raw data and the neighbourhoods. However, you can create your own embedding (PCA, ...) and plot neighbourhoods (by assigning neighbourhood coordinates to the coordinates of their anchor cells). <code>add_umap_coords()</code> adds the UMAP coordinates to the RawData data.frame, the Neighbourhoods data.frame and also to the Correlations data.frame</i>
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Description

Creates UMAP space for raw data and adds the UMAP-coordinates UMAP1/UMAP2 to RawData, Neighbourhoods and Correlations. run: `umapAddedLocoResult <- add_umap_coords(locoResult)` to add the UMAP coords to the loco result. `umapAddedLocoResult` can be used to plot the correlations in neighbourhoods on a UMAP. Details: To plot local correlations across neighbourhoods LoCo needs a space to plot those correlations in. One possibility is to create the UMAP space of the raw data. Within this UMAP space you can inspect original raw features and additionally assign UMAP coordinates to neighbourhoods by taking the central cell of all neighbourhoods (the anchor cell) and assigning the UMAP coordinates of this cell to their neighbourhood. After running `add_umap_coords()` you can run `plot_local_correlation_map()` to plot local correlations within this UMAP space. You do not have to use these umap coordinates but this function makes it easy to create the UMAP space for the raw data and the neighbourhoods. However, you can create your own embedding (PCA, ...) and plot neighbourhoods (by assigning neighbourhood coordinates to the coordinates of their anchor cells). `add_umap_coords()` adds the UMAP coordinates to the RawData data.frame, the Neighbourhoods data.frame and also to the Correlations data.frame

Usage

```
add_umap_coords (
  locoResult,
  n_pcs = 0,
  scale = TRUE,
  metric = "euclidean",
```

```

    seed = 7
)

```

Arguments

locoResult	the result of run_loco
n_pcs	the number of components for PCA (default 0 = skip PCA embedding). If n_pcs > 0 the function firstly decomposes the input matrix into PCs and runs umap on the PCs. In this case scale is set to FALSE, as the PCA already scales the data.
scale	z-score normalize feature counts before calculating UMAP coords (TRUE). This is ONLY considered when n_pcs is 0. If n_pcs > 0 the functions uses the first <n_pcs> PCs to run the UMAP embedding and does not z-score normalize in between!
metric	distance-metric for UMAP (euclidean)
seed	seed for UMAP (7)

```
plot_cell_level_correlation
```

Plot correlation between featureA and featureB as a scatter plot using all cells contained in neighbourhoods that are within given space boundaries [x_min ... x_max] and [y_min ... y_max].

Description

Plot correlation between featureA and featureB as a scatter plot using all cells contained in neighbourhoods that are within given space boundaries [x_min ... x_max] and [y_min ... y_max].

Usage

```

plot_cell_level_correlation(
  locoResult,
  featureA,
  featureB,
  x_min,
  x_max,
  y_min,
  y_max,
  dim1 = "UMAP1",
  dim2 = "UMAP2"
)

```

Arguments

locoResult	The UMAP-annotated result of run_loco
featureA	first feature (x-coordinate) of the plotted correlation

featureB	second feature (y-coordinate) of the plotted correlation
x_min	The minimum x-coordinate to filter neighbourhoods that are used for plotting. Cells in neighbourhoods are included if the anchor cell of this neighbourhood has an x-coordinate $\geq$ x_min.
x_max	The maximum x-coordinate to filter neighbourhoods that are used for plotting. Cells in neighbourhoods are included if the anchor cell of this neighbourhood has an x-coordinate $\leq$ x_max.
y_min	The minimum y-coordinate to filter neighbourhoods that are used for plotting. Cells in neighbourhoods are included if the anchor cell of this neighbourhood has an y-coordinate $\geq$ y_min.
y_max	The maximum y-coordinate to filter neighbourhoods that are used for plotting. Cells in neighbourhoods are included if the anchor cell of this neighbourhood has an y-coordinate $\leq$ y_max.
dim1	the x-axis dimension for the plot: per default the UMAP1 coordinate from add_umap_coords
dim2	the y-axis dimension for the plot: per default the UMAP2 coordinate from add_umap_coords

```
plot_local_correlation_map
```

Plot correlations for a correlationPair in all neighbourhoods. Therefore create a plotting space first (like UMAP coordinates) to then plot neighbourhoods into this space by, e.g., running

```
newLocoResult <- add_umap_coords(locoResult)
```

plot_local_correlation_map(newLocoResult, "GENE1_GENE2")

Description

Plot correlations for a correlationPair in all neighbourhoods. Therefore create a plotting space first (like UMAP coordinates) to then plot neighbourhoods into this space by, e.g., running `newLocoResult <- add_umap_coords(locoResult)` `plot_local_correlation_map(newLocoResult, "GENE1_GENE2")`

Usage

```
plot_local_correlation_map(
  locoResult,
  correlationPair,
  dim1 = "UMAP1",
  dim2 = "UMAP2"
)
```

Arguments

locoResult	the result of run_loco after adding an embedding space to the data.frame Correlations in the loco-result this can be generated by, e.g., running locoResult2 <- add_umap_coords(locoResult1) and then feeding locoResult2 into this function.
correlationPair	the name of the correlation that should be plotted. You can see all possible names in the column <FeaturePair> in <locoResult1\$LaplacianScores>.
dim1	the x-axis dimension for the plot: per default the UMAP1 coordinate from add_umap_coords
dim2	the y-axis dimension for the plot: per default the UMAP2 coordinate from add_umap_coords

run_loco	<i>Run LoCo analysis</i>
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Description

Run LoCo analysis

Usage

```
run_loco(
  inFile,
  del = "\t",
  col = FALSE,
  row = FALSE,
  zscore = TRUE,
  thread = 1,
  correlatedSetMode = 1,
  numberCorrelations = 0,
  cellStateGeneFile = "",
  correlationStateGeneFile = "",
  numberNeighbourhoods = 0,
  neighbourhoodSize = 100,
  neighbourhoodKNN = 5,
  correlationCutoff = 0.7,
  permutations = 100,
  minSetSize = 2,
  corrSetAbundance = 0.01
)
```

Arguments

inFile	Input file path.
del	Delimiter used in input file.

<code>col</code>	Whether feature names are in columns.
<code>row</code>	Whether cell names are in rows.
<code>zscore</code>	Apply z-score normalization.
<code>thread</code>	Number of threads.
<code>correlatedSetMode</code>	Mode for correlated sets (connected components, fully connected, connected by min x edges)
<code>numberCorrelations</code>	Number of correlations to compute.
<code>cellStateGeneFile</code>	Path to cell state gene file. Only these features are used to create cell neighbourhoods.
<code>correlationStateGeneFile</code>	Path to correlation state gene file: only these features are used to calculate pairwise correlations.
<code>numberNeighbourhoods</code>	number of neighbourhoods to create. Per default (0), LoCo creates $x = (\text{total-CellNumber} / \text{neighbourhoodSize})$ neighbourhoods.
<code>neighbourhoodSize</code>	Size of neighbourhoods.
<code>neighbourhoodKNN</code>	Number of nearest neighbours.
<code>correlationCutoff</code>	Correlation threshold (minimum correlation between pairs to consider this correlation)
<code>permutations</code>	Number of permutations for p-value calculation.
<code>minSetSize</code>	Minimum set size.
<code>corrSetAbundance</code>	Minimum abundance threshold (percentage of neighbourhoods that must contain a correlated pair to consider this pair).

Details

The result is a list of data.frame containing: 1.) the RawData, 2.) LaplacianScores: scored correlation pairs that vary across single-cell sapce, 3.) Correlations: all correlations in all neighbourhoods, 4.) Neighbourhoods: the definition of neighbourhoods by their anchor cell and all contained cells

Value

A named list containing LoCo results with four elements:

RawData A data.frame containing the input expression matrix in long form, with an additional column `cellID`. If the raw data did not contain named cellIDs the new cellIDs are enumerated in the form 'C_<index>', where the first index starts from 0. Rows correspond to cells and columns correspond to features.

LaplacianScores A data.frame summarising statistically significant feature pairs based on Laplacian scoring. Contains:

- **FeaturePair:** names of feature pairs as 'feature1_feature2'
- **LaplacianScore:** computed laplacian score
- **p_value:** permutation-based p-value
- **FeatureSet:** comma-separated list of features forming sets of co-correlated features

Correlations A data.frame of correlation values per neighbourhood with 3 columns "CorrelationPair", "NeighbourhoodID", "Correlation". The first column `CorrelationPair` contains feature pairs (same name as in `LaplacianScore`), the second column `NeighbourhoodID` contains all neighbourhoodIDs (like `N_<number>`), and the third column `Correlation` contains the correlation of the pair in this neighbourhood.

Neighbourhoods A data.frame describing all constructed neighbourhoods:

- **NeighborhoodID:** unique ID of each neighbourhood in the form 'N_<index>' where the index starts from 0.
- **AnchorCellID:** central cell of the neighbourhood. LoCo starts by sampling random cells (anchor cells) and builds local neighbourhoods around them.
- **AllCellIDs:** comma-separated list of all cells in this neighbourhood. These were the closest cells to the anchor cell.

Examples

```
# Load example dataset shipped with the package
infile <- system.file("example", "data_1.tsv",
                      package = "loco")

# Run LoCo on a small example dataset: find all local correlations with a correlation above 0.4
L1 <- run_loco("test/data_1.tsv", correlationCutoff= 0.4, neighbourhoodSize = 25)
# Inspect the correlations with the lowest laplacian score/ p_value -
# these correlations change the most across the whole single-cell dataset but change only a little
head(L1$LaplacianScores)
L2 <- add_umap_coords(L1)
plot <- plot_local_correlation_map(L2, L2$LaplacianScores$FeaturePair[1])
ggplot2::ggsave(filename = "local_correlation_map.png",
plot = plot,
width = 8,
height = 6,
dpi = 300)
```